# Band-1 Hadamard Order 9792 – Classical Paley-I (q=9791); IN SPK via Paley-I; Gap 2052 Still OPEN

H_9792 Paley-I PASS; Paley-II FAIL/SKIP (q=4895=5*11*89 NOT prime / not prime power; ===3 mod 4); Kronecker FAIL/SKIP (4*2448; 2448===0 mod 4 admissible but NOT in ONBOX_SEEDS); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

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**Document class.** Scientific computational / construction certificate (Band-1).
**Program.** Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
**Parents.** `2026-09-22_Band1_Hadamard_9788_PaleyI`; `2026-09-22_Band1_Hadamard_9768_PaleyI`; `2026-09-22_Band1_Hadamard_9744_PaleyI`.
**Lab.** `/workspace/experiments/band1/hadamard_9792_paleyI/`.
**Cutoff.** 2026-09-23T00:41:06Z (~8:41PM ET).

**Honesty stamp (non-negotiable).**
This memo claims **finite construction of order 9792 only (verified max|HH^T-nI|=0), via classical Paley-I q=9791 (prime ===3 mod 4). 9792 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 9792: Paley-I **PASS** ($q = 9791$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** ($q = 4895 = 5 \cdot 11 \cdot 89$ NOT prime / not prime power; $\equiv 3 \pmod 4$); Kronecker **FAIL/SKIP** (note $9792 = 4 \cdot 2448$; $2448 \equiv 0 \pmod 4$ Hadamard-admissible but NOT in ONBOX_SEEDS). **9792** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **9788+9768+9744+9740+9720+9680+9644+9632+9624+9620+9588+9552+9548+9540+9512+9492-** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

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## Abstract

Finite construction of order **9792**.

- $H_{9792}$ = Paley-I ($q = 9791$) – classical SPK

- Paley-II for 9792: **FAIL/SKIP** ($q = 4895 = 5 \cdot 11 \cdot 89$ NOT prime / not prime power; $\equiv 3 \pmod 4$)
- Kronecker for 9792: **FAIL/SKIP** ($4 \cdot 2448$; $2448 \equiv 0 \pmod 4$ Hadamard-admissible but NOT in ONBOX_SEEDS)
- Skip: Miyamoto digression this wave

**SPK: IN SPK via Paley-I**.
**Gap 2052:** OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

```
python3 verify_hadamard_9792.py
```

**Results.** overall_pass=True; units 155/155; row blocks 77/77 (block size 128); wall 755.0501s; constructed max|HHˆT-nI|=0.

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## Arithmetic

- Target $n = 9792 = q + 1$ with $q = 9791$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 4895 = 5 \cdot 11 \cdot 89$ NOT prime / not prime power; $\equiv 3 \pmod 4$ — FAIL/SKIP.
- Kronecker: $9792 = 4 \cdot 2448$; $2448 \equiv 0 \pmod 4$ Hadamard-admissible but NOT in ONBOX_SEEDS — FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

## Constructions (verified)

| Object | Method | max\|HHˆT-nI\| | Wall (s) |
|---|---|---|---|
| H_9792 Paley-I | Paley-I q=9791 | 0 | 746.5609 |
| H_9792 Paley-II | FAIL/SKIP (q=4895=5*11*89 NOT prime / not prime power; ===3 mod 4) | – | – |
| H_9792 Kronecker | FAIL/SKIP (4*2448; 2448===0 mod 4 admissible but NOT in ONBOX_SEEDS) | – | – |

## SPK closure note

Seeds include prior SPK through **9788** lineage (**9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; . . . ; **NOT** 8948/8644/8516/. . . ) + verified **9792** (+ prior **9788+9768+9744**). Next constructive gap **2052** (OPEN).

| Quantity | Value |
|---|---|
| 9788 in SPK | **True** (Paley-I q=9791) |
| Gap 2052 | **OPEN** |
| clay_closed | false |
| GATE-04 | LOCKED |

## Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

## Next optional

q=9803->n=9804 Paley-I DIRECT q=9803 OR q=9811->n=9812 Paley-I DIRECT q=9811 OR n=2052 standard SPK constructive gap – method-attack / lit OR q=9839->n=9840 Paley-I DIRECT q=9839 OR Track B / Object 1 meter await